

# SHIVAM KUMAR

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## Education

National Institute of Technology, Rourkela

August 2023 – Present

*Bachelor of Technology in Industrial Design Engineering (CGPA: 7.60/10)*

*Rourkela, Odisha*

## Relevant Coursework

- Data Structures and Algorithms
- Database Management Systems
- Web Development
- Object Oriented Programming
- Operating System
- Statistics & Probability

## Work Experience

Full Stack Development Intern (Emoji Encryption System)

May 2026 - Present

*Criticalsoft*

*Remote*

- Developed an emoji-based encryption protocol supporting **over  $10^{82}$**  cipher permutations, preventing brute-force attack vectors.
- Built the encryption backend using **React and Node.js**, reducing processing latency by **40%** during concurrent data requests.
- Implemented authentication using **JWT and Google OAuth**, establishing secure role-based access control across all user sessions.

## Projects

Autonomous Repository Security Auditor | [GitHub](#)

February 2026

*JavaScript, Node.js, OpenAI Agent SDK, GitHub API*

- Built a repository scanning tool using **JavaScript and the OpenAI Agents SDK** to identify leaked API credentials and authentication tokens.
- Implemented regex-based parsing and asynchronous recursive directory traversal, reducing scan times for large codebases by **35%**.
- Created an automated scoring pipeline to categorize security risks and suggest remediation steps, cutting manual code review time by **45%**.

Social Network Friend Suggestion System | [GitHub](#)

October 2025

*C++, OOPS, Data Structures*

- Created a serialized graph data structure using adjacency lists to map and query **100+ nodes and 500+ relational edges**.
- Executed Breadth-First Search (BFS) algorithms to compute second-degree interpersonal connections, reducing search time by **25%**.
- Applied priority queue mechanisms via max-heaps to rank connection viability, improving recommendation accuracy by **30%** based on mutual network overlap.

## Technical Skills

- **Languages:** C++, Python, JavaScript, SQL, HTML, CSS
- **Frameworks & Libraries:** Node.js, React.js, Express.js, Bootstrap
- **Databases:** MySQL, MongoDB
- **Tools & Platforms:** Git, GitHub, VS Code, REST APIs

## Achievements & Certifications

- **micro1 Certified Full Stack Developer:** Cleared rigorous AI-driven technical interview, demonstrating advanced Full Stack Development competencies. {Certificate} **May 2026**
- **Certified MERN Stack Developer – Apna College:** 120+ hrs mastering React, Node.js, Express, MongoDB, APIs, and scalable deployment. {Certificate} **June 2025**

## Extracurricular Activities

- **Management Head – Intra Hall Events, VS Hall, NIT Rourkela:** Directed a cross-functional team of **15+ members** to organize collegiate events and manage logistics. **January 2024**